

Loops Advanced

Learning Objectives

- Use repeat, forever, and repeat-until loops in Scratch
- Understand when to use different loop types
- Create animated sequences using loops effectively

Engage (5-7 minutes)

In a village, daily routines repeat: wake up, tend cattle, go to field, return home. Loops in Scratch automate repetition — like telling a computer to repeat an action many times without writing it again and again, just as a farmer’s daily schedule repeats throughout the season.

Explore (10-12 minutes)

Students experiment with three loop types: “repeat” for a set number of times, “forever” for endless repetition, and “repeat-until” for condition-based repetition. Like a spinning potter’s wheel, loops create continuous motion and action in their programs.

Explain (10-12 minutes)

The “repeat” block executes code a specified number of times. The “forever” block runs indefinitely until stopped. The “repeat-until” block continues until a condition becomes true. Like the cyclical nature of seasons — planting, growing, harvesting — loops create patterns that repeat with purpose.

Elaborate (8-10 minutes)

Students create a spinning animation using a repeat loop that changes sprite direction. Like a water wheel turning continuously, their sprite moves in repeated patterns, demonstrating how loops create efficient, elegant code.

Evaluate (5-8 minutes)

Can students differentiate between repeat, forever, and repeat-until loops? Can they create an animation that loops at least 10 times? Do they understand how loops reduce code repetition?